

EARTH AND TIMBER

As human societies developed agriculture and former nomads began to settle, temporary encampments gave way to permanent structures, perhaps as far back as 10000 BCE. The earliest building material was the earth itself, combined with other materials, such as straw, to add strength, and timber to support roofs.

Building with earth

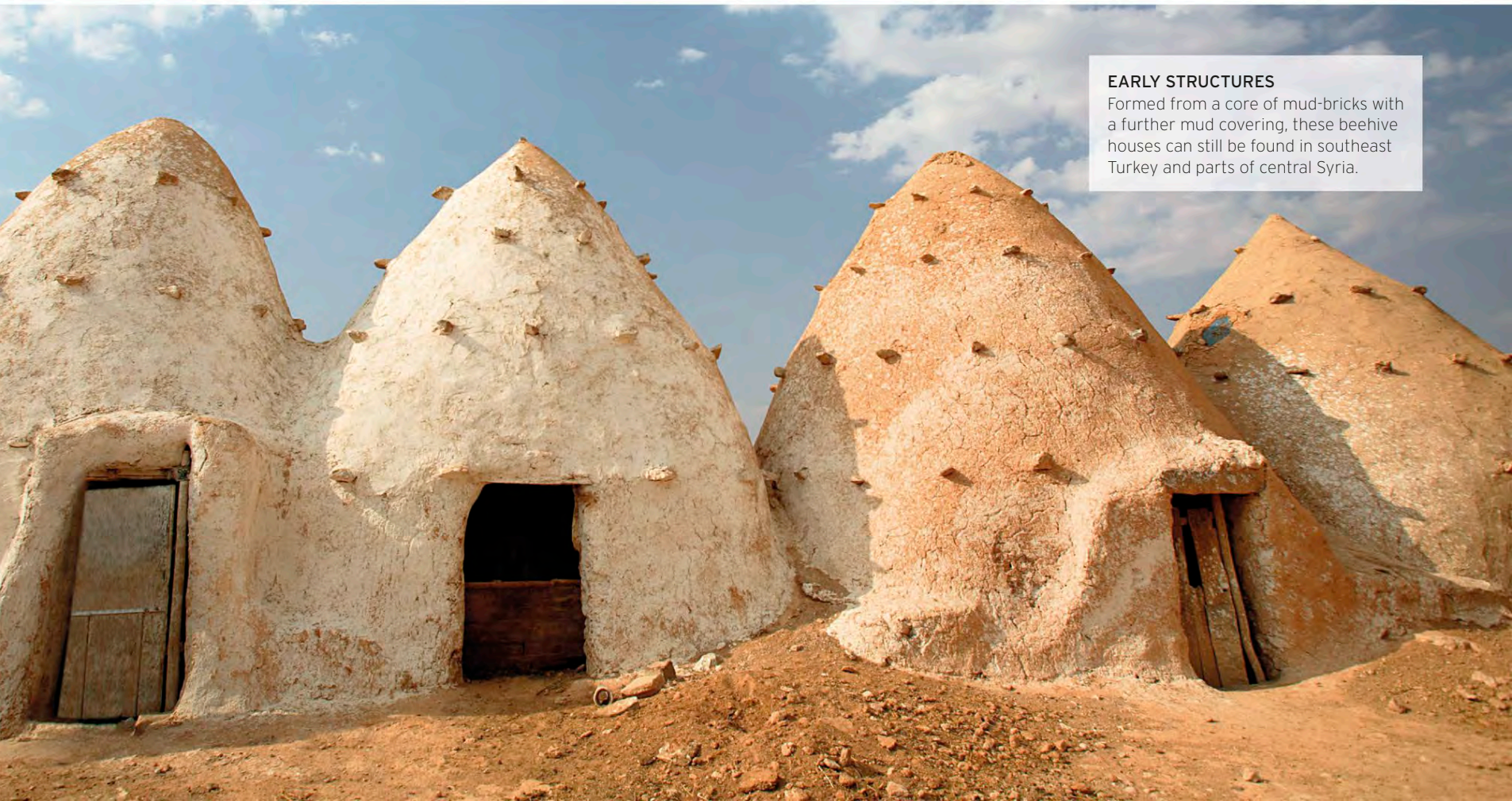
Earthen structures are highly susceptible to erosion and weathering, and little evidence of early human earthen dwellings remains today. The structures that do survive have done so largely because of their massive size. These tend to be barrows, mounds, and tumuli; the names tend to be interchangeable, but they are all huge, man-made hills of compacted earth of a size to defy the elements. Such mounds are found across northern and central Europe and on the steppes of Russia and Central Asia. Many date from the Bronze Age (c.3500–300 BCE). Their function is almost always as a tomb, and archaeologists have discovered skeletons and funerary items in many of them. One of the largest mounds in Europe is the Leeberg near Großmugl in Austria, which dates from c.600–500 BCE. This is dwarfed by the burial mound at Bin Tepe in Lydia, Turkey, built in around 560 BCE; with a height of 196 ft (60 m), it is one of the largest tumuli known.

Mud and mud-mixture structures

Some of the earliest habitable buildings were made from mixtures of mud or dung strengthened with straw or horse hair, sometimes applied to a frame of branches or woven reeds. It does not seem a large step from building with mud and straw in bulk to building with bricks made of mud and straw, but the compressive strength and versatility of fired bricks allowed humans to build sizeable and sophisticated structures. The earliest large-scale buildings of mud-brick, or adobe, belong to the civilizations of ancient Mesopotamia (modern-day Iraq). Built in around 2100 BCE, the Ziggurat of Ur (see p.243) was a huge stepped pyramid that is estimated to have risen to over 98 ft (30 m). Later Mesopotamian civilizations developed glazed bricks, which they used to decorate their buildings, as at Babylon, their capital.

Earth remains one of the most abundant basic materials. It is low-tech, easily worked, durable, fire resistant, and cheap, so mud-brick is still popular for building—although not in regions facing regular rainfall.

The laminated **timber** used in modern high-rise buildings is similar in **strength** to **steel** but much **lighter**



EARLY STRUCTURES

Formed from a core of mud-bricks with a further mud covering, these beehive houses can still be found in southeast Turkey and parts of central Syria.